Profile Coherence as a Diagnostic Lens for Financial Asset Recommendation

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Abstract

Financial asset recommendation (FAR) literature optimises ranking quality and realised return, but no published metric exposes whether recommendations are profile-aligned in the regulatory sense MiFID II suitability requires. We introduce Profile Coherence at k (PC@k), scoring the share of a top-k list whose asset risk class lies within ordinal tolerance of the customer's declared MiFID II risk band. Applying PC@k to FAR-Trans [Sanz-Cruzado *et al.*, 2024], we find that 18.6% of 228,241 scoreable Buys violate the declared band by two or more bands, with a sharply bimodal per-customer self-discordance distribution. Yet a transaction-level Ordinary Least Squares (OLS) regression on 208,029 priced Buys, with cluster-robust standard errors on `customerID`, finds coherent Buys earn $+2.94$ percentage points (pp) higher 6-month realised return than discordant ones, conditional on volatility, segment, and year. We then propose a LightGCN extension that stratifies by band and adds a profile-coherent margin loss atop Bayesian Personalised Ranking (BPR), lifting aggregate PC@10 from 0.794 to 0.918 on all data splits, at no return on investment cost.

1 Introduction

Financial asset recommendation (FAR) sits at the intersection of personalised ranking and regulatory obligation. The European Markets in Financial Instruments Directive (MiFID II) requires firms to assess suitability before recommending a product: alignment between the product's risk profile and the customer's declared risk tolerance, investment horizon, and loss capacity. A recommender that maximises predicted return without conditioning on the customer's risk band can be an excellent ranker on standard benchmarks but a regulatory liability.

FAR-Trans [Sanz-Cruzado *et al.*, 2024] is the first open FAR benchmark with both customer profile metadata and a multi-year transaction history: 228,913 Buys across 806 assets and 29,090 retail customers, with MiFID II risk bands per customer and metadata sufficient to derive each asset's risk class. The accompanying paper benchmarks LightGCN, Random Forest, and others on normalised Discounted Cumulative Gain (nDCG) and Return on Investment (ROI), each evaluated on the top-10 ranked items. We will additionally report Recall alongside these as a ranking sanity check. None of those metrics measures profile alignment, and none of the published baselines conditions on the declared risk band.

This paper introduces a profile-alignment lens for FAR and applies it in four steps: *diagnose* the prevalence of discordance, *test* whether it carries an economic penalty, *audit* where existing baselines sit on the new axis, and *remedy* the gap with a profile-coherent training objective. These map onto four research questions:

- **RQ1 (Diagnose)**: How prevalent is profile-discordance in observed FAR-Trans Buy transactions, and is it a stable customer trait or transaction-level noise?
- **RQ2 (Test the economic objection)**: Do profile-discordant Buys earn lower realised six-month return than coherent ones, once asset volatility, customer segment, and year are controlled for?
- **RQ3 (Audit existing baselines)**: At their best published settings, do the chosen FAR-Trans baseline models lift profile coherence above a naive band-aware reference (one that picks within the customer's declared band) for every band?
- **RQ4 (Remedy)**: Can a stratified, profile-coherent LightGCN improve profile coherence over the global baseline, and at what cost to nDCG, Recall, and ROI?

Contributions. This paper makes two contributions. First, we introduce Profile Coherence at top-k ranked items (PC@k), a coherence metric with a band-conditional random baseline $\pi(b)$ and scale-invariant lift normalisation. Second, we propose a stratified, profile-coherent LightGCN with a margin loss atop Bayesian Personalised Ranking (BPR), together with an ablation that isolates the coherence-loss effect from the stratified architecture.

2 Related Work

2.1 FAR benchmarks and the two-axis evaluation status quo

FAR-Trans [Sanz-Cruzado *et al.*, 2024] is the first public FAR benchmark with both real retail-investor transactions and asset pricing, and it sets the evaluation tone for the field: 11 baselines are reported on nDCG@10 for next-purchase ranking quality and ROI@10 for the realised return. The follow-up of Sanz-Cruzado *et al.* [2026] shows that these two axes are not aligned on FAR-Trans: transaction-based and profitability-based metrics are negatively correlated, and models that win one lose the other. However neither axis measures whether a recommendation aligns with the customer's declared MiFID II risk band, which is the gap PC@k addresses.

2.2 Profile-aware FAR architectures

Ghiye *et al.* [2024] extend LightGCN with a causal graph convolution and a sliding-window training scheme for credit-bond recommendation, capturing temporal drift in user interests via embedding updates restricted to recent interactions. Their result supports the broader point that LightGCN can be modified to carry domain-specific structure beyond plain BPR ranking, and motivates the choice of LightGCN as the backbone for our stratified, profile-coherent extension.

2.3 Risk-aware FAR via post-hoc re-ranking

Sakurai *et al.* [2026] propose a risk-aware utility re-ranking (RURA) procedure on FAR-Trans: a mean-variance utility conditioned on the customer's MiFID risk tolerance is applied to a base ranker's top list, lifting ROI@10 while preserving nDCG@10. RURA optimises a risk-tolerance utility but does not quantify how aligned the resulting list is with the declared band on an ordinal-distance basis. PC@k is complementary: it is a metric for that alignment, and is computable on any ranker (with or without re-ranking).

All in all, existing FAR work optimises profitability or ranking, with risk-tolerance entering only as a post-hoc utility weight. No prior FAR evaluation metric operationalises MiFID II suitability as an ordinal coherence score over top-k lists. This paper closes that gap with PC@k and a profile-coherent training-time objective.

3 Methodology

3.1 Profile Coherence

Ordinal band scale

Customers and assets share the same 4-point ordinal scale, running from Conservative (0) through Income (1) and Balanced (2) to Aggressive (3). The customer band b_u is taken from FAR-Trans's `riskLevel` field, with a flag distinguishing declared values from regression-imputed `Predicted_*` ones. Customers whose entry is `Not_Available` or missing are excluded. The asset band b_i follows the hierarchical rule in Section 4.2.

Pairwise discordance

$$d(u,i) = |b_u - b_i| \in \{0,1,2,3\}. \tag{1}$$

A recommendation is *profile-coherent* iff $d \leq 1$. The strict variant $d = 0$ appears only as a sensitivity row.

Aggregation

$$\text{PC@}k(u) = \frac{1}{k}\left|\{i \in \text{top}_k(u) : d(u,i) \leq 1\}\right|. \tag{2}$$

Two implementation rules close the definition. First, recommenders returning fewer than k items are scored on what they returned. Second, customers with $b_u = \emptyset$ are dropped from the aggregate. Per-split PC@k is the unweighted mean across eligible customers and the headline numerics average over all splits.

Band-conditional random baseline

The closed-form expectation of PC@k under a uniformly random recommender for band b is given by

$$\pi(b) = \frac{|\{i \in A : |b - b_i| \leq 1\}|}{|A|}, \tag{3}$$

where A is the asset universe. Section 4.2 reports the FAR-Trans asset-band distribution and the resulting $\pi(b)$ values.

PC-lift@k

Per-user PC@k is normalised against the random reference at the customer's own band:

$$\text{PC-lift@}k(u) = \frac{\text{PC@}k(u)}{\pi(b_u)}. \tag{4}$$

Lift > 1 beats the band-conditional random rate, while lift $= 1$ matches it. Because 77.9% of FAR-Trans assets sit in the Conservative, Income, and Balanced bands, a recommender that pushes Income assets at everyone would score well on raw PC@k for any user whose tolerance set includes Income. Dividing by $\pi(b_u)$ strips the band-specific easiness of the asset universe and makes per-band values commensurate.

3.2 Stratified, Profile-Coherent LightGCN

Backbone

LightGCN is the backbone: a differentiable loss with per-customer embeddings lets us attach the coherence margin to the BPR objective. Random Forest is not extended as its objective uses asset-level technical indicators only and carries no customer signal.

Stratified architecture

We train four band-stratified sub-models, one per declared MiFID risk band. At inference, each customer is routed to the sub-model for their declared band.

Profile-coherent margin loss

BPR samples a positive item from observed Buys and a negative item uniformly. We add a margin term pushing the score of a discordant draw below that of a coherent draw:

$$\mathcal{L}_{\text{coh}}(u) = \text{softplus}(s(u,i_d) - s(u,i_c)), \tag{5}$$

Here $s(u,i)$ is the LightGCN inner-product score between customer u and asset i. A coherent draw i_c satisfies $d(u,i_c) \leq 1$ and a discordant draw i_d satisfies $d(u,i_d) > 1$.

Total loss

$$\mathcal{L} = \mathcal{L}_{\text{BPR}} + \omega\|\Theta\|_2^2 + \lambda \cdot \mathcal{L}_{\text{coh}}, \tag{6}$$

with ω the L2 weight decay on embedding parameters Θ and mixing weight $\lambda \geq 0$. BPR remains the primary ranking signal and only the treatment trial activates the coherence term.

Ablation Setup

$\lambda = 0$ leaves stratified BPR alone, isolating the architecture's contribution, while $\lambda = 1$ adds the coherence margin. The two trials decompose the full intervention into stratification and PC-loss pieces.

4 Experimental Setup

4.1 Dataset

Beyond the headline counts already noted in Section 1, FAR-Trans [Sanz-Cruzado *et al.*, 2024] additionally provides 159,136 Sell transactions, with the 806 assets split across stocks, bonds, and mutual funds, and daily closing prices spanning January 2018 to November 2022. The MiFID II `riskLevel` field used throughout this paper takes values in {Conservative, Income, Balanced, Aggressive}, with `Predicted_*` variants flagging regression-imputed bands. In addition, customer profiles also carry `customerType` and `investmentCapacity`, which enter the RQ2 controls in Section 4.5.

4.2 Asset Risk Band Assignment

Each asset is assigned to one of the four MiFID risk bands by a three-step hierarchical rule, applied in order.

Step 1: subcategory metadata

Mutual funds are labelled directly from their subcategory: Money Market funds are treated as Conservative, Bonds as Income, Balanced as Balanced, and Equity or Large Cap as Aggressive. Bonds are labelled by issuer type: Government as Conservative, Corporate as Income, and any remaining bond subcategory as Income by default.

Step 2: volatility quartiles for stocks

Stocks carry no subcategory tag we can use, so they fall through to a volatility rule. For each ISIN we compute the annualised standard deviation of daily log returns on a trailing 252-day window, then bucket all stocks into the four bands by the stock-only quartile thresholds (q_1, q_2, q_3).

Step 3: residual default

Any asset that falls through both prior steps is assigned to Balanced.

The resulting asset-band distribution across (Conservative, Income, Balanced, Aggressive) is $(190, 333, 105, 178)$ over $|A| = 806$ assets, which yields a band-conditional random baseline of $\pi(b) = (0.65, 0.78, 0.76, 0.35)$ under the formula defined in Section 3.1.

4.3 Temporal Split Protocol

Evaluation uses 69 monthly temporal splits on a trading-day grid between August 2019 and April 2025. For each split, training is all Buys before `time_point`; test is the window $(\texttt{time\_point}, \texttt{test\_end}]$, deduplicated against training and filtered to the training-asset universe. Eligible customers have at least one interaction in both training and test, while eligible assets have prices on both endpoints. Per-split deltas are paired against the global LightGCN baseline.

4.4 Baselines and Hyperparameter Grids

Table 1 reports the search grids and fixed values for the two FAR-Trans baselines. Both grids are swept with Ray Tune, with LightGCN tuned on average nDCG@10 and Random Forest on average ROI@10 across the 69 splits. Best-trial values are reported in the appendix.

Parameter	Value	Role
LightGCN (tuned on average nDCG@10)		
`embedding_dimension`	$\{64, 128\}$	grid
`number_of_layers`	$\{2, 3\}$	grid
`learning_rate`	$\{10^{-2}, 10^{-3}\}$	grid
`weight_decay`	10^{-5}	fixed
`keep_probability`	0.6	fixed
`number_of_epochs`	50	fixed
`batch_size`	1024	fixed
Random Forest (tuned on average ROI@10)		
`number_of_estimators`	$\{20, 30, 40, 50\}$	grid
`max_depth`	$\{15, 25, 50\}$	grid
horizon	6 months	fixed
`random_state`	42	fixed

Table 1: Baseline hyperparameter grids. "grid" rows are searched, "fixed" rows are held constant.

The stratified PC-LightGCN inherits the best LightGCN trial's backbone hyperparameters. Two trials are evaluated, $\lambda = 0$ (BPR only, isolating the stratified architecture) and $\lambda = 1$ (coherence margin activated), on the same 69 splits.

4.5 Statistical Inference

Three of the four research questions from Section 1 require statistical analysis beyond the headline metrics. RQ1 is descriptive and needs no inference test. The remaining three map to the following procedures.

- **RQ2: Does discordance carry a return penalty?** Transaction-level OLS on 208,029 priced Buys (Equation 7), with cluster-robust SE on `customerID` and two-sided Wald at $\alpha = 0.05$. Each customer contributes about 8 Buys on average, so within-customer residuals share unobserved shocks and motivate the clustered errors.
- **RQ3: Do baselines lift coherence on every band?** Per-customer PC@10 is averaged unweighted over (customer, split) within each (declared band, model) cell and per-band lift is the ratio of that mean to $\pi(b)$.
- **RQ4: Does PC-LightGCN improve coherence, and at what cost?** Aggregate metrics are averaged over the 69 splits, paired per-split deltas are computed against the baseline LightGCN, and the win rate is the share of splits on which the treatment beats the comparator.

The RQ2 OLS specification is

$$\begin{aligned} r = {} & \beta_0 + \beta_1 \cdot \text{is\_coherent} \\ & + \beta_2 \cdot \text{asset\_volatility} \\ & + \mathbf{C}(\text{type})\boldsymbol{\gamma} \\ & + \mathbf{C}(\text{year})\boldsymbol{\delta} \\ & + \varepsilon. \end{aligned} \tag{7}$$

where $r = (p_{\text{end}} - p_{\text{start}})/p_{\text{start}}$ is the 6-month realised return, trade-date and end prices are matched by backward-asof (7-day tolerance, rows over a week dropped), and `is_coherent` is binary.

4.6 Reproducibility

All code, configuration grids, and evaluation artefacts are available in the project repository at https://github.com/Samthesimpsons/SMU-Capstone.

5 Results and Analysis

5.1 RQ1: Profile-Discordance is Prevalent and Structural

We characterise discordance in four slices, starting from the pooled transaction rate and drilling into the customer, band, and year dimensions in turn. At the transaction level (Figure 2), of 228,241 scoreable Buys:

1. 18.6% violate the declared band by two or more steps;
2. 81.4% are coherent, of which 34.1% strictly ($d = 0$);
3. mean discordance is 0.874 bands, so the typical violation is a one-band miss.

The pooled rate cannot tell us whether these violations are i.i.d. transaction-level slips spread thinly across the customer base or a structural trait concentrated in a subset. Slicing by customer settles the question. The per-customer discordant share (Figure 1) is sharply bimodal at 0% and 100% rather than clustering near the 19% mean. More explicitly, 64.4% of banded customers are fully coherent across every Buy and 17.2% are fully discordant. An i.i.d. noise process cannot produce this shape, making discordance a persistent customer-level trait.

If the trait is customer-level, the next question is which customers carry it. Coherence per declared band is U-shaped (Figure 3): Conservative at 45.1% (reach for yield), Aggressive at 56.2% (regression toward safer assets), with the two centre bands at ∼90%. The two ordinal extremes carry most of the mismatch while the centre bands are nearly fully coherent, which corroborates the bimodality finding from the customer slice.

A final slice rules out a regime-driven artefact. Mean discordance is flat across 2019 to 2022 (within 0.02 bands; Figure 4), so the pattern is not the product of any single year's market conditions. The 2018 value (0.98) sits higher but covers a small slice. MiFID II came into force on 2018-01-03, the exact start of FAR-Trans, so we hypothesise the elevation as a transition effect. However this hypothesis is not tested and is potential future work.

5.2 RQ2: Coherent Buys Earn a Conditional Return Premium

The model from Section 4.5 is fit on 208,029 priced Buys with cluster-robust SE on `customerID`. Table 2 reports the full coefficient table.

Coherent Buys earn $+2.94$ pp higher 6-month realised return than discordant ones, conditional on volatility, segment, and year. The positive sign with $p < 10^{-12}$ observationally falsifies the prior that regulatory coherence imposes a return tax. The raw slice gap is $+3.31$ pp, so controls absorb only ∼0.4 pp (∼11% of the unconditional gap). This implies that coherence is not a volatility, segment, or year confound. However, it is important to note that unobserved confounders (financial literacy, advisor channel, account constraints) are not ruled out.

The volatility coefficient is -10.3 pp per unit, consistent with low-volatility outperformance over 2019 to 2022 and the 2022 rate shock that hit high-beta names. All four year dummies reject H_0: 2020 strongly positive (COVID rebound) and 2022 negative (Fed-hike bond drawdown). All four `customerType` dummies fail to reject H_0: segmentation contributes no detectable return signal, and together with RQ1's bimodality it does not capture the persistent customer-level discordance trait either.

5.3 RQ3: Both Baselines Under-Serve at Least One Band

At primary-metric optima (Table 3), LightGCN wins ranking quality (nDCG 0.330, Recall 0.495, PC@10 0.794) but loses ROI (-0.5%/mo). On the flip side, Random Forest wins ROI ($+1.4\%$/mo) but ranks near random on nDCG (0.019) and Recall (0.036). Both look fine on aggregate PC (LightGCN PC-lift $1.17\times$, Random Forest $1.06\times$), but the aggregate hides per-band failures.

Model	nDCG@10	ROI@10	Recall@10	PC@10
LightGCN	0.330	−0.005	0.495	0.794
RF	0.019	+0.014	0.036	0.667

Table 3: RQ3 aggregate metrics at primary-metric optima, averaged over 69 splits. ROI@10 is per-month.

Table 4 decomposes PC@10 by declared band. Per-band lift moves $0.75 \rightarrow 1.13 \rightarrow 1.17 \rightarrow 1.35$ for LightGCN and $0.52 \rightarrow 0.70 \rightarrow 1.03 \rightarrow 1.88$ for Random Forest across Conservative → Income → Balanced → Aggressive. For both, lift rises monotonically with the declared band: recommended assets skew higher-risk. LightGCN's range is flatter, which we attribute to its per-user embeddings shifting each customer's top-k partly toward their tolerance window. Recall that Random Forest carries no customer signal.

Band	Model	N	$\pi(b)$	PC@10	PC-lift@10
Conservative	RF	5,348	0.65	0.34	0.52
Conservative	LightGCN	5,348	0.65	0.48	0.75
Income	RF	45,142	0.78	0.54	0.70
Income	LightGCN	45,142	0.78	0.88	1.13
Balanced	RF	61,181	0.76	0.79	1.03
Balanced	LightGCN	61,181	0.76	0.89	1.17
Aggressive	RF	25,367	0.35	0.66	1.88
Aggressive	LightGCN	25,367	0.35	0.47	1.35

Table 4: Per-band PC@10 and PC-lift@10 versus the band-conditional random baseline $\pi(b)$ for the two FAR-Trans baselines at their primary-metric optima, aggregated over 69 splits. PC-lift@10 = PC@10 / $\pi(b)$.

By customer footprint, LightGCN under-serves the 19.7% Conservative segment while Random Forest under-serves 60.9% (Conservative + Income). Aggressive's high lift is partly mechanical: $\pi(\text{Aggressive}) = 0.35$ is the lowest band,

Term	Estimate	Std. Error	95% CI	p
Intercept	-0.0471	0.0141	$[-0.0746,\ -0.0195]$	8.1×10^{-4}
`is_coherent`	$+0.0294$	0.0040	$[+0.0215,\ +0.0372]$	2.3×10^{-13}
`asset_volatility`	-0.1025	0.0165	$[-0.1349,\ -0.0701]$	5.5×10^{-10}
`C(customer_type)[Legal Entity]`	$+0.0154$	0.0486	$[-0.0798,\ +0.1106]$	0.751
`C(customer_type)[Mass]`	$+0.0011$	0.0132	$[-0.0248,\ +0.0269]$	0.936
`C(customer_type)[Premium]`	$+0.0058$	0.0135	$[-0.0206,\ +0.0322]$	0.667
`C(customer_type)[Professional]`	$+0.0247$	0.0169	$[-0.0084,\ +0.0578]$	0.143
`C(year)[2019]`	$+0.1759$	0.0129	$[+0.1507,\ +0.2012]$	1.4×10^{-42}
`C(year)[2020]`	$+0.2107$	0.0061	$[+0.1986,\ +0.2227]$	6.2×10^{-259}
`C(year)[2021]`	$+0.0754$	0.0053	$[+0.0651,\ +0.0858]$	4.1×10^{-46}
`C(year)[2022]`	-0.0195	0.0045	$[-0.0283,\ -0.0107]$	1.4×10^{-5}

Table 2: Full RQ2 OLS coefficient table on 208,029 priced Buys, cluster-robust SE on `customerID`. Reference cell: Inactive customer, 2018, `asset_volatility` $= 0$, `is_coherent` $= 0$.

so any model skewing higher-volatility clears that bar easily. Neither baseline has band labels or a band-aware loss. As such, per-band coherence is a passive byproduct of training, not a controlled output.

5.4 RQ4: Stratification + PC-Loss Closes the Per-Band Deficit

Table 5 reports aggregate metrics for the baseline LightGCN, the $\lambda = 0$ ablation, and the $\lambda = 1$ treatment over 69 splits. The treatment ($\lambda = 1$) wins PC@10 universally. The aggregate metrics show PC@10 climbing from 0.794 to 0.918 ($+12.4$ pp) and PC-lift from $1.17\times$ to $1.40\times$, and the per-split win-rate plot (Figure 6) shows 100% on both across the 69 splits.

The gain decomposes cleanly into a stratification piece and a PC-loss piece, both read off the paired per-split deltas in Figure 5. Stratification alone ($\lambda = 0$ vs baseline) contributes $+2.7$ pp PC@10 ($\sim$22% of the full gain), with $+0.006$ Recall and $+0.002$ nDCG that the win-rate plot confirms as noise-level (72% and 57% wins respectively, Figure 6). Adding the PC-loss ($\lambda = 1$ vs $\lambda = 0$) buys another $+9.7$ pp PC@10 ($\sim$78%) at $\sim$2 pp nDCG and $\sim$3 pp Recall, with PC@10 won and nDCG / Recall lost on every split (Figure 6). Stratification supplies the cheap quarter while the PC-loss supplies the rest.

Aggregate ROI is essentially flat. Reading the ROI row of Table 5 together with the paired ROI deltas in Figure 5: $\lambda = 1$ vs baseline is -0.002 pp/mo, $\lambda = 0$ vs baseline is -0.002 pp/mo, and $\lambda = 1$ vs $\lambda = 0$ is $+0.0001$ pp/mo. The corresponding per-split ROI win rates (Figure 6) are 33%, 28%, and 58% respectively. The small ROI cost therefore lives in the stratification step, not in the coherence loss.

Together with RQ2's coherence premium, this null aggregate-ROI effect indicates that the recommender can be made meaningfully more profile-coherent without giving up the return story, reframing coherence as an architectural choice rather than a post-hoc tradeoff: compliance becomes a training-time objective expressible as an ordinal distance on a customer-asset metric pair.

6 Limitations

- *Observational nature.* The RQ2 coherence premium is associational as unobserved confounders (financial literacy, advisor channel, account constraints, prior portfolio composition) are not ruled out by the volatility, segment, or year controls. The elevated 2018 mean discordance is hypothesised as a MiFID II transition effect on similar grounds, but not directly tested against a pre-MiFID counterfactual.
- *Metric and band-assignment design choices.* The tolerance $d \leq 1$ is set rather than derived, binarising a 4-point ordinal distance. The asset-band assignment in Section 4.2 is a hierarchical heuristic not externally validated.
- *Intervention scope and tuning.* Only LightGCN is extended, with sequential, transformer, and content-based recommenders untested. Within LightGCN, $\lambda = 1$ is one point on a Pareto frontier between PC and ranking-quality metrics, and the intervention inherits the RQ3 best-trial backbone hyperparameters rather than being re-tuned end-to-end. The reported gains are therefore a lower bound on what a full frontier sweep or end-to-end re-tuning could achieve.
- *Generalisation and measurement scope.* Results are on FAR-Trans under MiFID II's 4-band scale, so generalisation to other jurisdictions, longer windows, or alternative asset universes is not established. Additionally, RQ2 measures realised return on executed Buys while RQ4 measures forward return on top-k recommended slates that may never be acted on, so the two ROI quantities are not directly comparable.

7 Conclusion

This paper frames **profile coherence** as a diagnostic lens for the FAR-Trans recommendation problem, and assembles the four research questions into a single diagnostic-to-remedy chain:

- **RQ1** establishes that the diagnostic axis is real and structural at the customer level, not transaction-level noise.
- **RQ2** removes the economic objection to optimising for coherence: the regulatory return tax prior is observationally falsified.
- **RQ3** locates the existing FAR-Trans baselines on the coherence axis and shows that both under-serve specific

Trial	nDCG@10	ROI@10	Recall@10	PC@10	PC-lift@10
Baseline LightGCN	0.330	−0.005	0.495	0.794	1.17×
$\lambda = 0$	0.332	−0.007	0.501	0.821	1.19×
$\lambda = 1$	0.309	−0.007	0.467	0.918	1.40×

Table 5: RQ4 aggregate metrics, averaged over 69 splits. ROI@10 is per-month. PC-lift@10 is PC@10 divided by the band-conditional random baseline $\pi(b)$, expressed as a multiple.

declared bands in ways the aggregate metrics hide.

- **RQ4** demonstrates that a stratified, profile-coherent LightGCN extension closes the per-band coverage deficit at no aggregate ROI cost, with the gain decomposing cleanly into a free stratification component and a bounded-cost PC-loss component.

The headline contribution is that the regulatory objective (MiFID-aligned band coherence) and the economic objective (realised return) are not in structural conflict on FAR-Trans: a recommender can be made meaningfully more profile-coherent while keeping the return story the data already supports. This reframes coherence as an architectural choice rather than a tradeoff to be managed post-hoc, treating compliance not as a post-hoc filter applied downstream of a ranking model but as a training-time objective expressible as an ordinal distance on a customer-asset metric pair. The stratification plus margin-loss template then generalises naturally to other suitability-style constraints (concentration limits, leverage caps, ESG bands) that share the same ordinal structure.

References

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A Supplementary Figures

The figures below visualise the per-customer, per-band, per-year, and per-split diagnostics referenced from the main text.

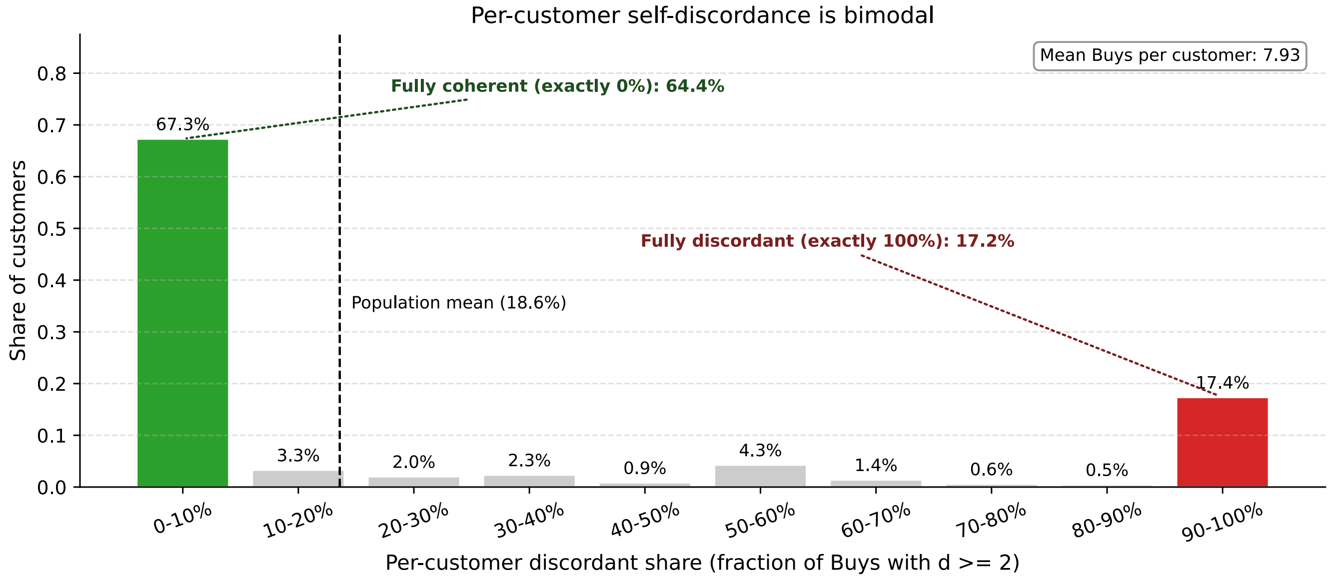

Figure 1: Per-customer share of discordant Buys is sharply bimodal at 0% and 100%, ruling out i.i.d. transaction-level noise.

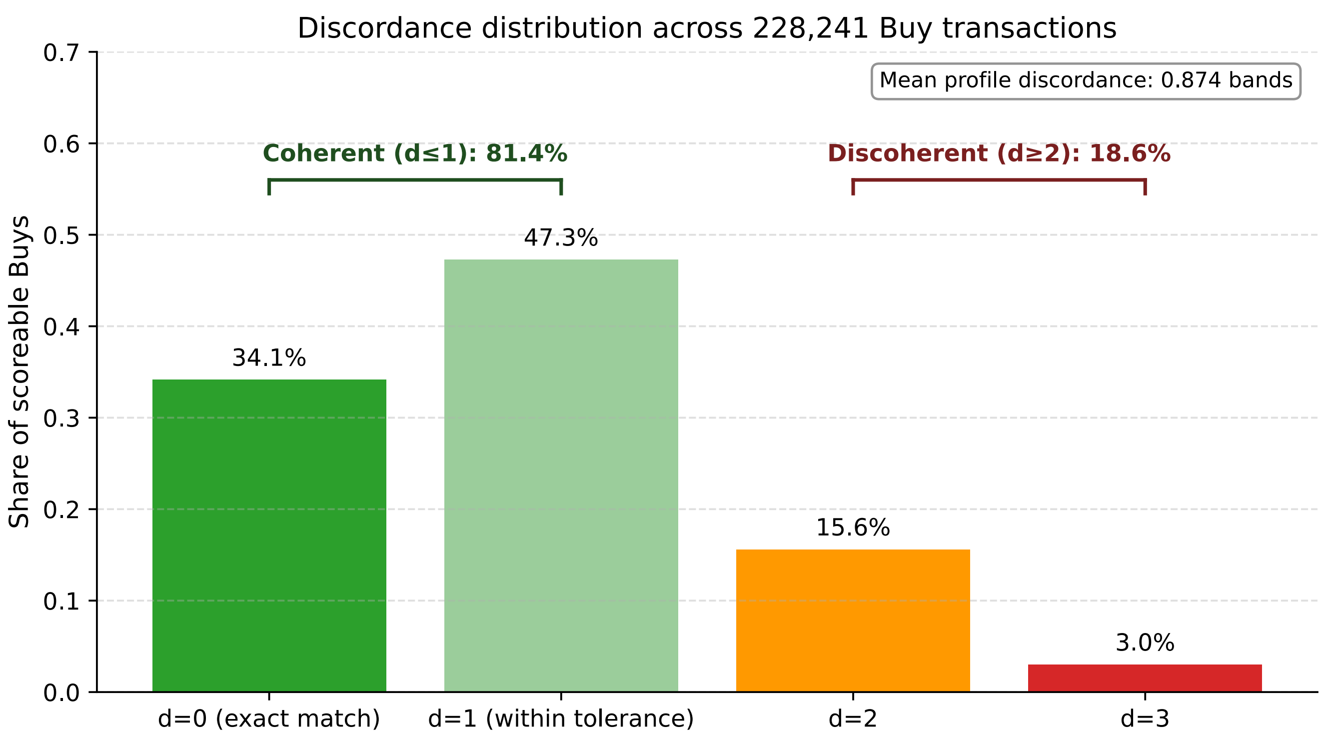

Figure 2: Distribution of pairwise discordance d over 228,241 scoreable Buys. Heavy-tailed shape; the typical violation is a one-band miss.

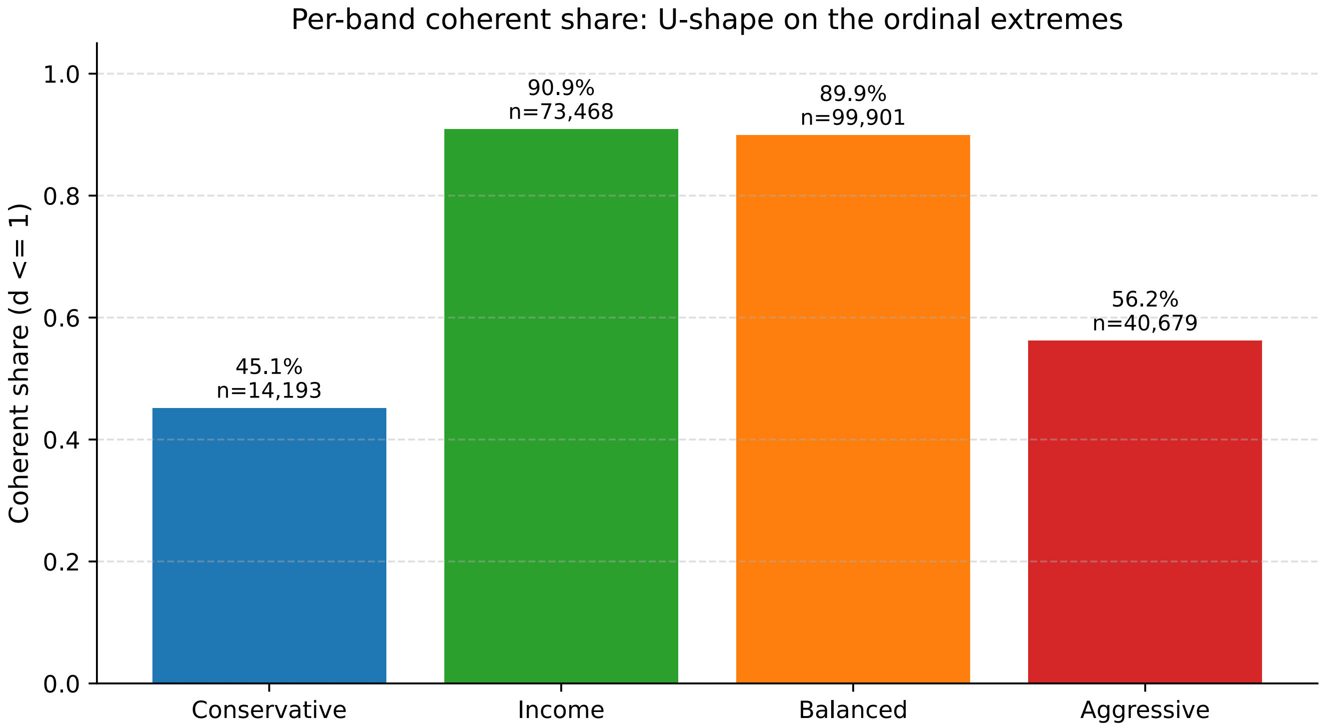

Figure 3: Coherence share by declared band. The U-shape supports the bimodality result of Figure 1: discordance concentrates on the ordinal extremes.

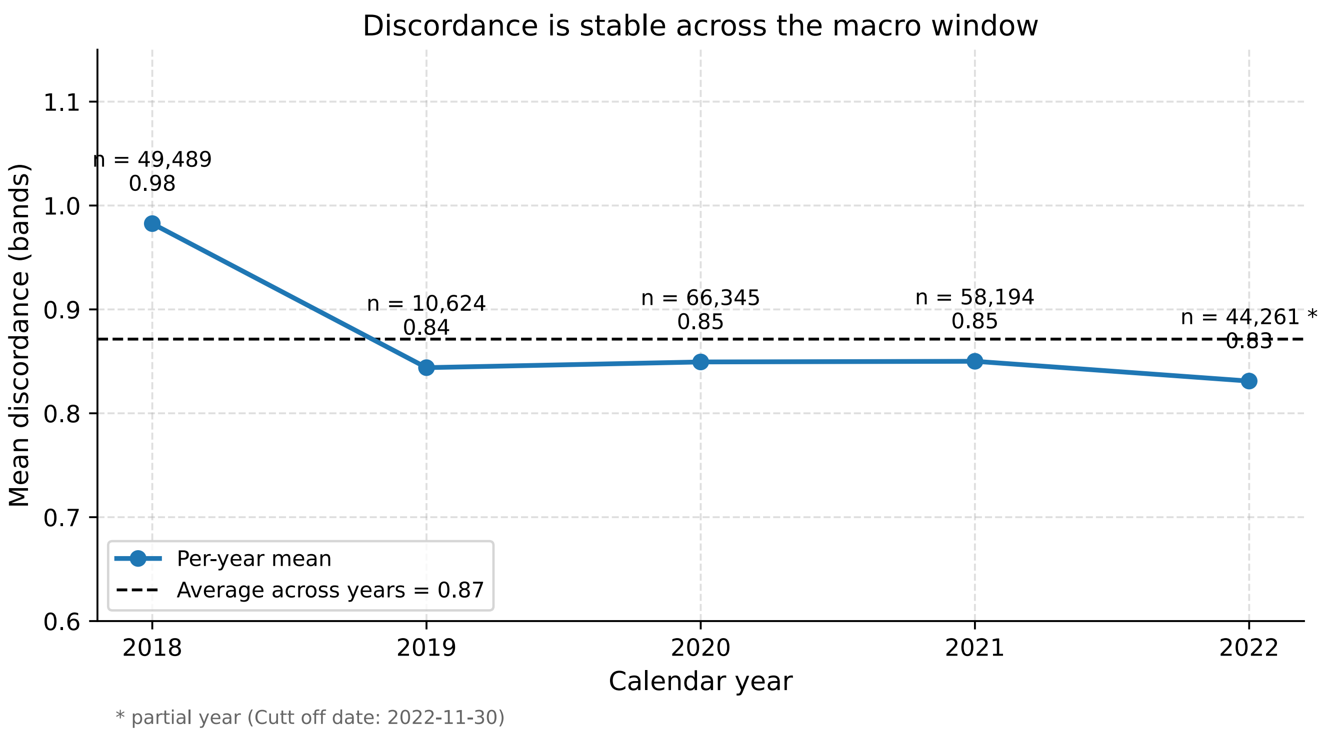

Figure 4: Mean discordance by calendar year. The 2019 to 2022 cluster sits within 0.02 bands; 2018 is elevated (transition-effect hypothesis).

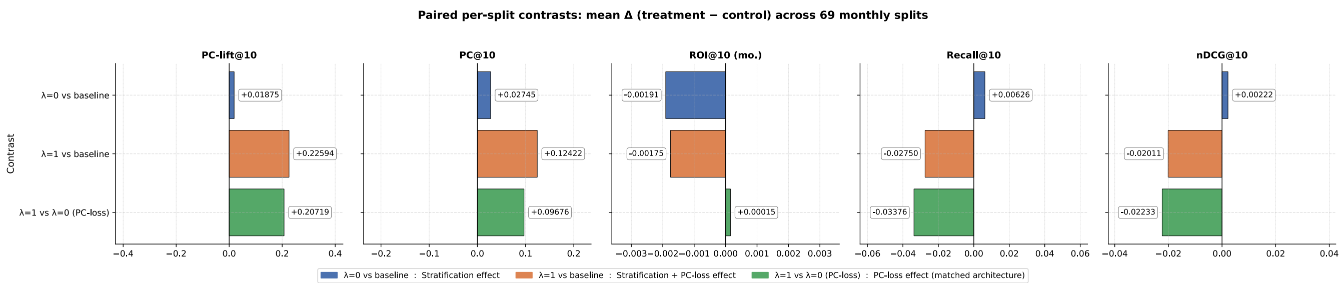

Figure 5: Paired per-split deltas of $\lambda = 1$ vs baseline LightGCN. PC@10 is positive on every one of 69 splits.

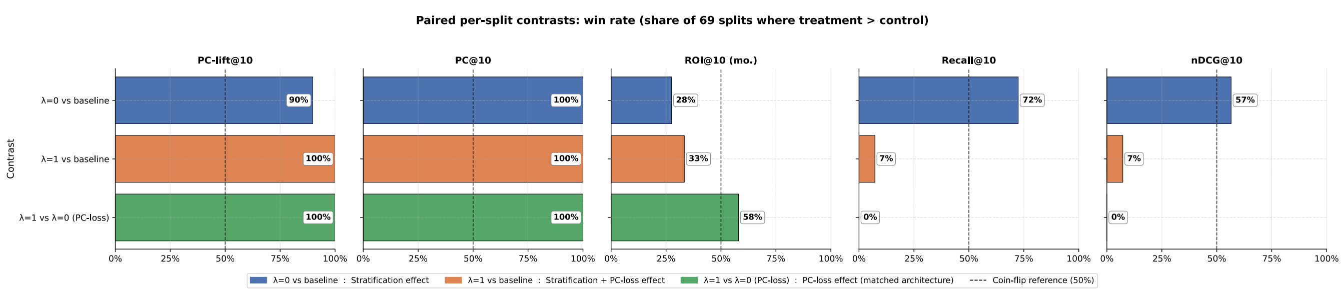

Figure 6: Per-split win rates of the $\lambda = 0$ and $\lambda = 1$ trials against the global LightGCN baseline.